

had high growth rates. The prices of these firms are bid up too high, reflecting excessive optimism. When this growth does not materialize, prices fall, leading to returns on growth stocks being low relative to value firms. The story here is that value stocks are not fundamentally riskier than growth firms, as in the rational stories. Value stocks are cheap because investors underestimate their growth prospects. Conversely growth firms are expensive because investors overestimate their growth prospects.

The value effect can also be produced by investors with other psychological biases. Barberis and Huang (2001) generate a value effect by employing two psychological biases: *loss aversion* and *mental accounting*. Loss aversion we have seen before, in chapter 2. Since investors suffer from losses more than they rejoice from equivalent gains, a loss following a loss is more painful than just a single loss. As for mental accounting, here agents look at each stock individually rather than considering overall gains and losses on their portfolios. The Barberis–Huang story of value is that a high book-to-market ratio stock is one that has achieved its relatively low price as the result of some dismal prior performance. This burned the investor who now views it as riskier and thus requires higher average returns to hold the stock.

The crucial question that behavioral models raise is: why don't more investors buy value stocks and, in doing so, push up their prices and remove the value premium, just as investors appear to have done with the size premium (at least according to the Grossman–Stiglitz interpretation)? Put another way, why aren't there more value investors? It can't be ignorance; the message of Graham and Dodd has spread far and wide since the 1930s, as demonstrated by the cult-like fervor of Berkshire Hathaway annual meetings, where value guru Warren Buffett holds court, or at the Graham & Dodd Breakfast conferences at Columbia Business School.

Perhaps investors think value investing is too difficult. Yet simple strategies of academics sorting stocks on a book-to-market basis are available even to the smallest retail investor using stock screens freely available on the Internet. Perhaps it is the legacy of the efficient market theory developed in the 1970s (see chapter 6)—but active managers have never believed in truly efficient markets, and now academics no longer believe in them either.

Maybe not enough institutions have sufficiently long horizons to effectively practice value investing. The value effect documented here, though, is different from the “deep value” practiced by some investors, including Buffett. That requires five- to ten-year horizons. The book-to-market value effect described here is a three- to six-month effect. But perhaps even this horizon is too long for most “long-horizon” investors.

Behavioral Implications for Asset Owners

The relevant question that an asset owner should ask from a behavioral standpoint is simple: do you act like the market, or do you have the ability not to